

Andrew Pfennigwerth

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Education

UNIVERSITY OF COLORADO BOULDER

Graduated December 2025

Bachelor of Arts in Computer Science, Minor in Business Analytics

Boulder, CO

- Chancellor's Achievement Scholarship Recipient
- Foreign Exchange Program in Vienna, Austria
- Relevant coursework: *Object-Oriented Analysis and Design, Software Development Methods and Tools, Database Systems, Introduction to AI, Applied Machine Learning, Financial Analytics*

Work Experience

Colorado Center for Astrodynamics Research

May 2025 – August 2025

Research Software Engineer

Boulder, CO

- Developed a Java-based Android plugin for the Android Team Awareness Kit (ATAK), a tactical mapping platform used by military and federal agencies, as part of a U.S. military-sponsored research project focusing on GPS integrity.
- Built a multi-source position overlay that ingests live GNSS, network, and fused location streams and renders them on the map, letting field operators detect spoofing by flagging divergence between sources.
- Implemented real-time signal graphs tracking metrics across satellite constellations to surface jamming spikes at a glance.
- Designed a color-coded integrity dashboard displaying AGC, SNR, and timing data so users can assess spoofing and jamming risk without interpreting raw GNSS data.
- Added a map-integrated signal health indicator that changes color based on live integrity metrics, giving non-expert operators an instant read on whether their navigation data can be trusted.

SciQuel

May 2024 – August 2024

Front End Engineer Intern

Remote

- Designed and iterated on UI prototypes in Figma in collaboration with the engineering team, focusing on accessibility and ease of use.
- Owned front-end development of the article submission and editing flow, taking the feature from design to production.
- Built and maintained React components across the codebase, contributing to a consistent and responsive user interface.

Projects

FlyStream

AI-Powered Fly Fishing Tool

- Built an end-to-end LangChain agent that takes a location, pulls live fishing reports, then cross-references results against a curated regional fly pattern database to generate grounded, location-aware recommendations.
- Replaced an earlier vector search approach (ChromaDB + MiniLM embeddings) with lightweight in-memory ranking after profiling memory constraints on free-tier deployment, preserving recommendation quality while improving speed and cost.
- Incorporated structured Tavily API calls and a curated fallback dataset, eliminating hallucinated river names while maintaining reliability when live data is unavailable.
- Shipped full-stack: FastAPI backend (Render), React frontend (Vercel); unit-tested extraction, ranking, and fallback behavior.

Quizletics

Full-stack trivia web application with real-time leaderboard

- Designed and implemented PostgreSQL database schema for user profiles, game sessions, and leaderboard rankings.
- Built RESTful API endpoints in Node.js to fetch trivia questions from external API and handle game logic.
- Developed front-end interface with JavaScript, managing game state and dynamic score updates.
- Collaborated with 4 teammates using Agile methodologies and Git workflow

Skills

- **Languages:** Python, C++, Java, Javascript, HTML, CSS, SQL
- **Frameworks & Libraries:** Node.js, React, Pandas, Android SDK
- **Devops & Tools:** Git, Docker, Cursor, Claude Code, Figma
- **Core Ideas & Practices:** Data Structures, Algorithms, REST APIs, Agile Methodology, LLM Integration